Lab Task#8 (A)

Sec A – In Lab

Note: All statements should be commented. Plagiarism will get straight zero.

This task is done in the lab. Use the `with` statement for every file you open. Do not import any module except `os`. The data file is given in the `data` folder: keep this folder next to your `.py` files, run your programs from that folder, and build every path with `os.path.join`. Submit all your files in one zipped folder.

```
Q1.py   Q2.py

data/

    marks.csv    (given)
```

Question 1

The marks of a class are stored in `data/marks.csv`. The first line is the header:

```
Roll No,Name,Quiz,Assignment,Mid,Final
```

The quiz and the assignment are out of 10, the mid is out of 30 and the final is out of 50.

1) Read the file into a list of rows and convert the marks of every student to `int`.

2) Write `grade(total)` that returns "A" if total ≥ 85, "B" if total ≥ 70, "C" if total ≥ 50, otherwise "F". Include a docstring.

3) For every student print the roll number, the name, the total out of 100 and the grade.

4) Print the class average (2 decimal places) and the topper(s) with their total (more than one student may have the highest total).

5) Write a new file `data/result.csv` with the header `Roll No,Name,Total,Grade` and one line per student. Use `join` to build each line. Read the file back and print it (you can also open it in Excel).

Question 2

A student keeps a diary of daily expenses in a file. The expenses of this week are:

```
expenses = [("01-Oct", "food", 450), ("01-Oct", "transport", 120),
            ("02-Oct", "books", 900), ("03-Oct", "food", 380),
            ("04-Oct", "transport", 150), ("05-Oct", "food", 520)]
```

1) Write `add_expense(filename, date, category, amount)` that adds one expense at the end of the file as a line `date,category,amount`. Use append mode.

2) Call `add_expense` for every expense, with the file `data/expenses.txt`.

3) Write `read_expenses(filename)` that returns the expenses saved in the file as a list of `(date, category, amount)` tuples, with the amount as an `int`. Print the list.

4) Print the total spent, a dictionary with the total of every category, and the category with the highest spending.

5) Run your program twice. What happens to the total the second time, and why? Explain in a comment, then fix the program by opening the file once in `'w'` mode at the start.

Include a docstring in each function.

Hints:

• Every line read from a file ends with `'\n'`. Use `line.strip()` before you split or compare it.

• Everything read from a file is a string: convert it with `int()` or `float()` before doing arithmetic, and convert numbers back with `str()` before writing them.

• `','.join(["a", "b", "c"])` gives `"a,b,c"`; every item of the list must be a string.

• Opening a file in `'w'` mode erases it, even if you write nothing:

```
with open(filename, "w") as f:
    pass
```